

Loan Default & Credit Risk Analysis

Data Analytics Mini Project

Prepared by: Vishnu Burusu

1. Project Objective

The objective of this project is to analyze loan and customer information to identify patterns associated with loan default. The analysis focuses on demographics, income, credit score, loan amount, interest rate, debt-to-income ratio, previous defaults, employment, and other credit-risk indicators.

2. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI Desktop, Power Query and DAX.

3. Dataset Description

The cleaned dataset contains 2,000 records and 28 columns. Important fields include Customer_ID, Annual_Income, Credit_Score, Loan_Type, Loan_Amount, Interest_Rate, Debt_to_Income_Ratio, Previous_Defaults, Region and Default_Status.

4. Data Cleaning & Feature Engineering

Duplicate records were removed. Missing numeric values in Annual_Income, Credit_Score and Employment_Years were filled using medians; missing Property_Ownership values were filled using the mode. Text values were trimmed, numerical outliers were checked with the IQR method, and Default_Flag, Credit_Score_Group, Income_Group and DTI_Group were created.

5. Key Analysis Results

Metric	Value
Total Customers	2,000
Total Loans	2,000
Defaulted Loans	224
Non-Defaulted Loans	1,776
Overall Default Rate	11.20%
Average Annual Income	745,402.55
Average Credit Score	691.98
Average Loan Amount	1,337,582.50
Average Interest Rate	9.44%
Average DTI	47.18%

6. Python Visualizations

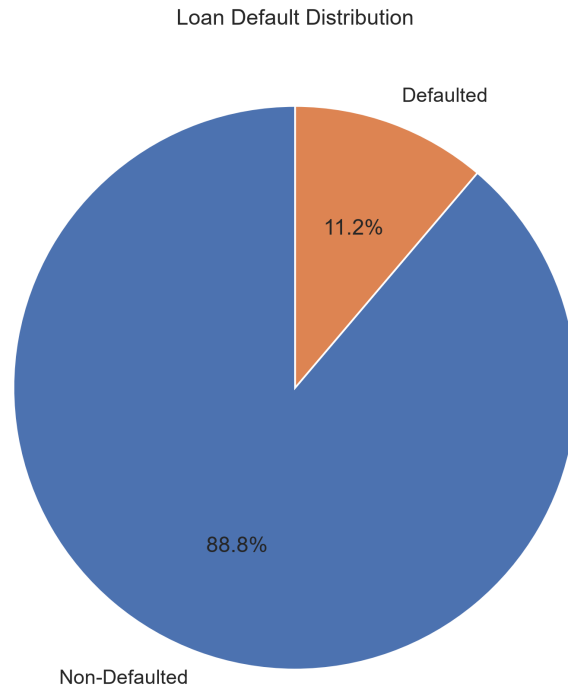


Figure 1. Loan Default Distribution

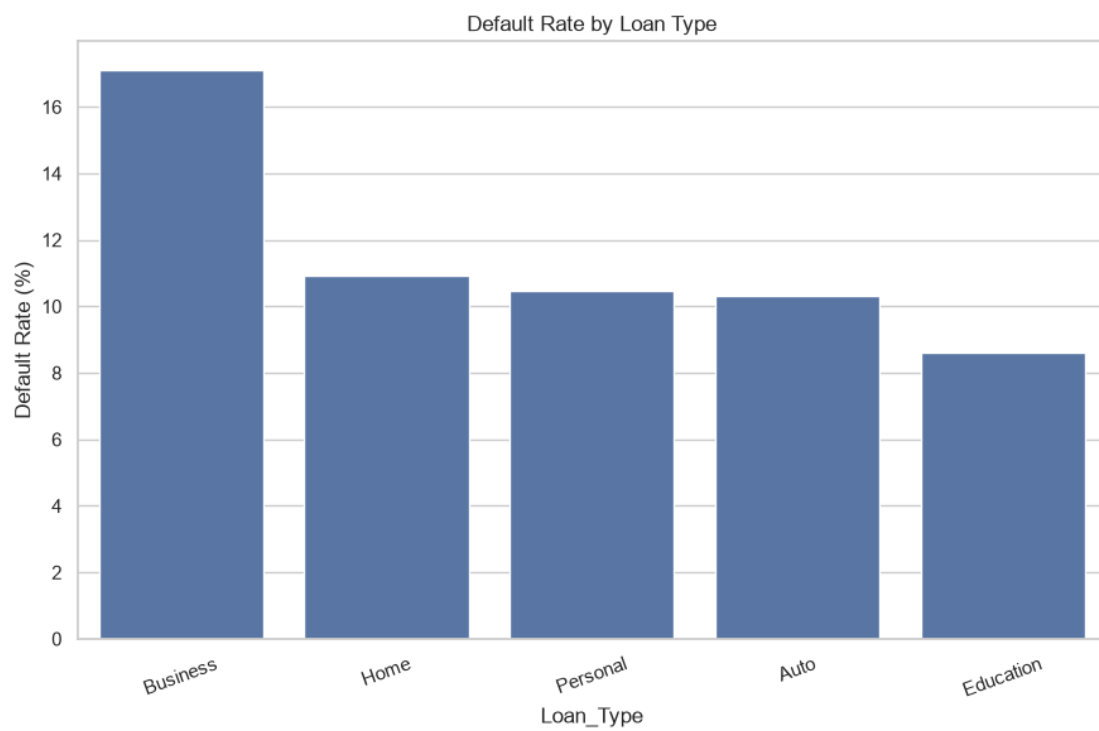


Figure 2. Default Rate by Loan Type

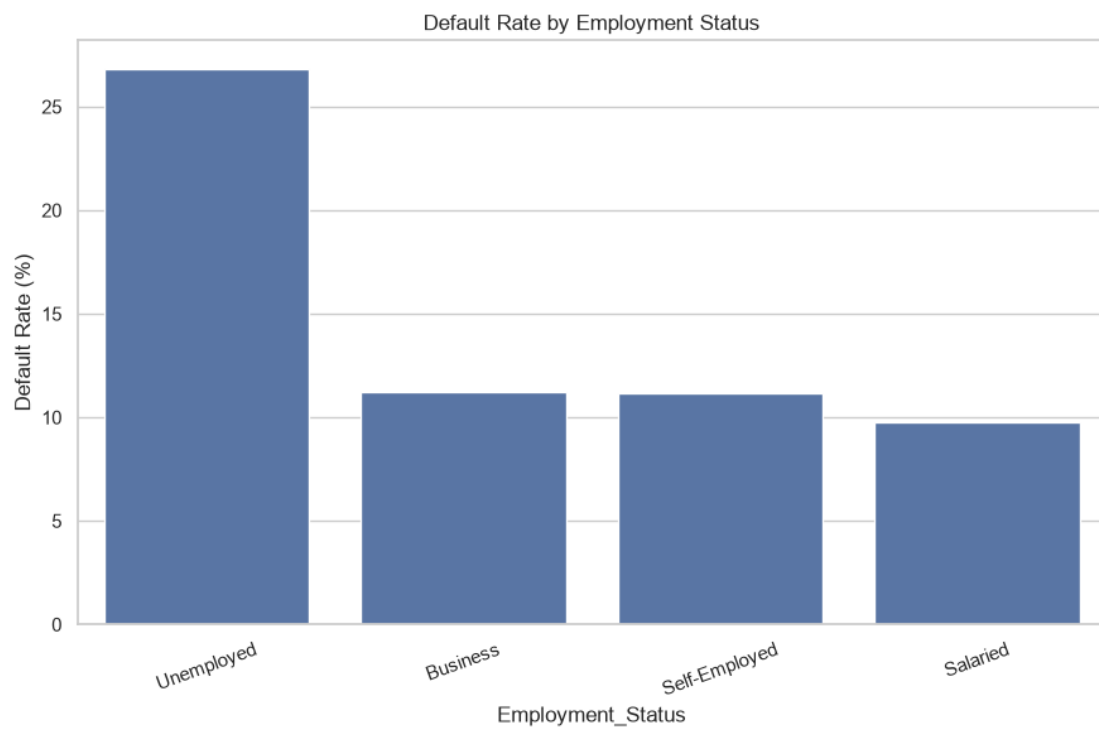


Figure 3. Default Rate by Employment Status

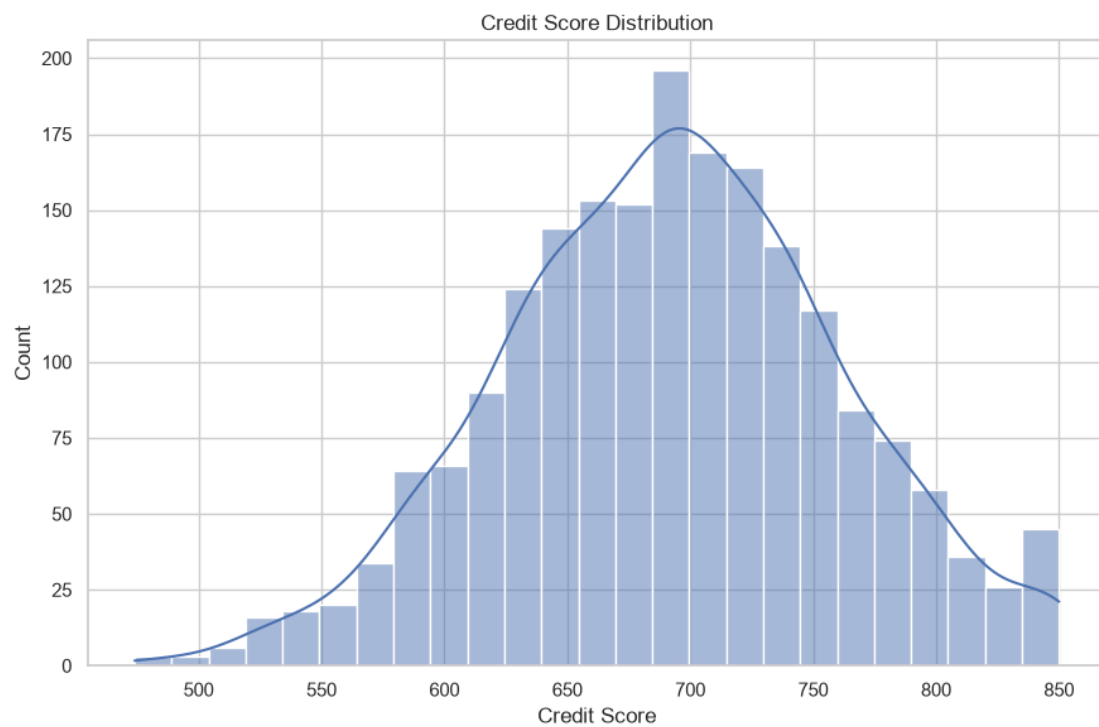


Figure 4. Credit Score Distribution

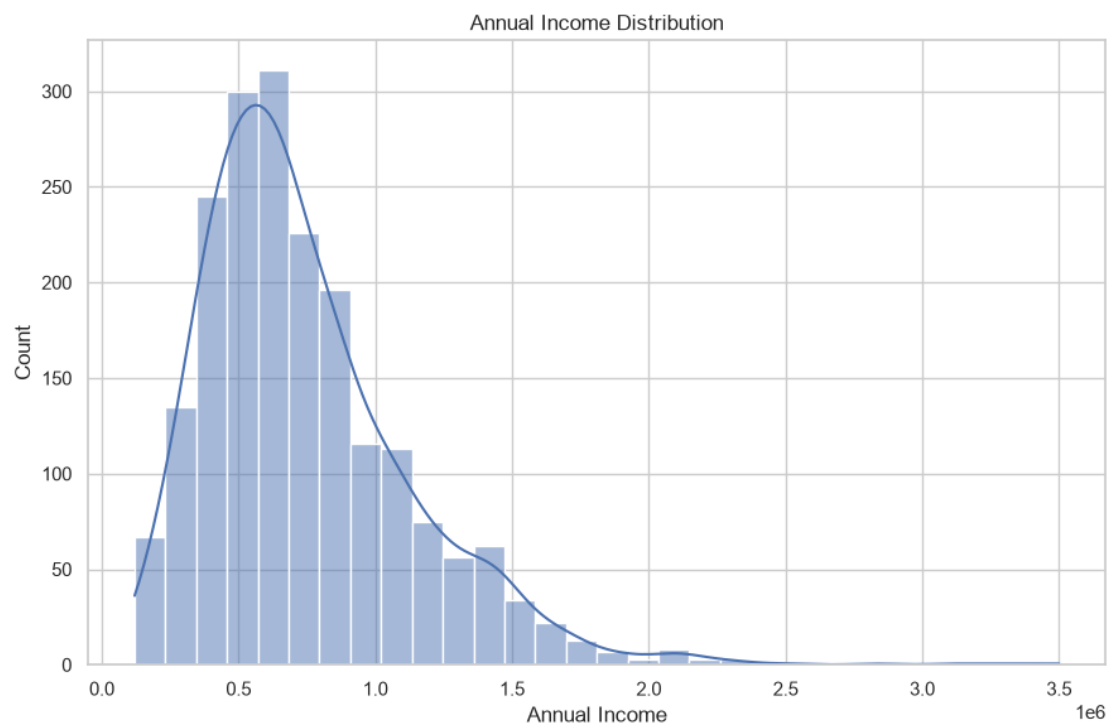


Figure 5. Annual Income Distribution

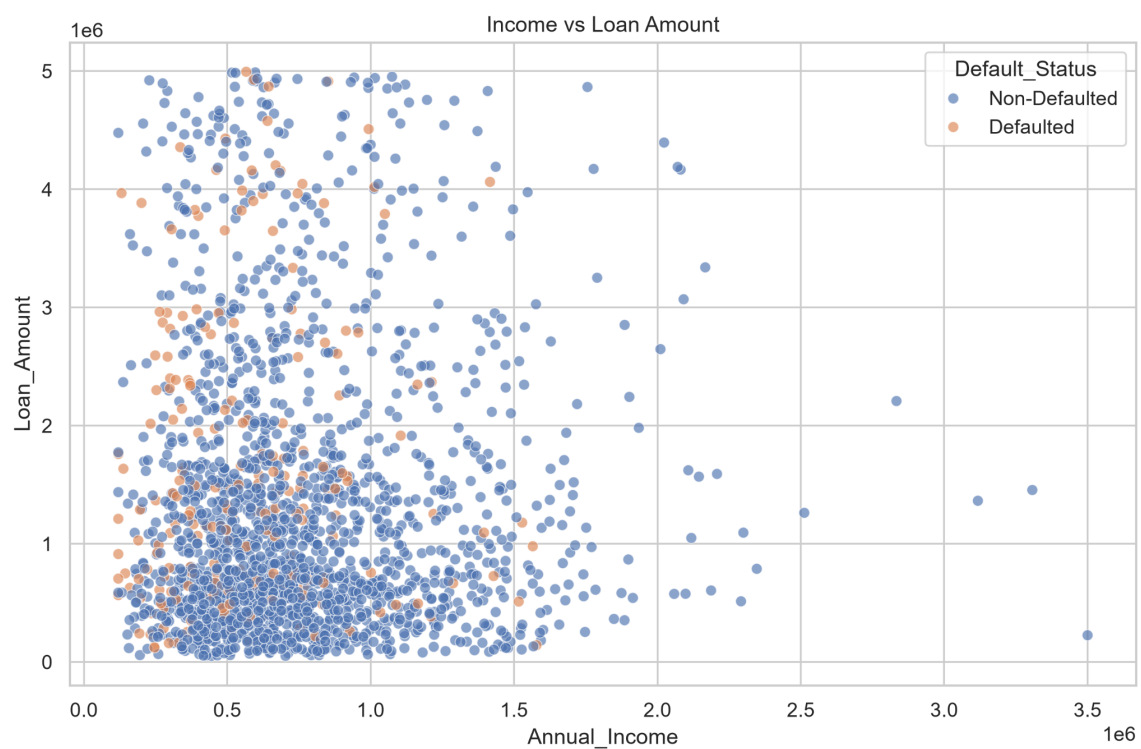


Figure 6. Income vs Loan Amount

7. Business Insights

1. Overall loan default rate is 11.20%, meaning about 11.2 out of every 100 loans in the dataset are defaulted.
2. Business loans have the highest default rate (17.11%).
3. South records the highest regional default rate (13.29%).
4. Unemployed borrowers show the highest default rate by employment status (26.85%).
5. Average credit score is lower among defaulted borrowers (654.36) than non-defaulted borrowers (696.73).
6. Average DTI is higher among defaulted borrowers (63.43%) than non-defaulted borrowers (45.14%).
7. Borrowers with 3 previous defaults have the highest current default rate (63.64%).
8. The highest-risk sufficiently sized segment is Poor credit / Low Income (75.00% default rate).
9. Average loan amount for defaulted borrowers is 1,621,553.57, compared with 1,301,766.33 for non-defaulted borrowers.
10. Credit score, DTI, previous defaults, income, employment status and loan type should be monitored together rather than relying on a single risk indicator.

8. Power BI Dashboard

The Power BI component is designed as a single-page interactive dashboard using KPI cards, default-risk comparisons, credit-score and income segmentation, a loan-details table, and slicers. Required DAX measures include Total Customers, Total Loans, Defaulted Loans, Non-Defaulted Loans, Default Rate %, Average Loan Amount, Average Credit Score and Average Annual Income.

9. Conclusion

The project demonstrates an end-to-end data analytics workflow: raw-data loading, cleaning, feature engineering, exploratory analysis, visualization and interactive dashboarding. The analysis indicates that default risk varies meaningfully across credit quality, debt burden, previous-default history, employment status and loan categories. These indicators can support ongoing credit-risk monitoring and portfolio analysis.